

Gene Conversion Homogenization and Lewontin's Paradox of Variation

A Comprehensive Research Document

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1. Introduction and Background

1.1 The Puzzle

At the heart of population genetics lies a fundamental prediction: under neutral theory, genetic diversity should scale linearly with effective population size. The standard neutral model (Kimura 1969; Ewens 2004) predicts that equilibrium diversity at neutral sites is $\pi = 4N_e\mu$, where N_e is the effective population size and μ is the per-site mutation rate per generation.

Yet empirical data tell a very different story. Across metazoan species, census population sizes vary over **many orders of magnitude**—from a few hundred individuals in endangered species to billions in insects and marine organisms (Lewontin 1974; Charlesworth 2009). If the neutral prediction were correct, we should observe diversity varying by many orders of magnitude as well. Instead, observed neutral diversity varies by only **about two orders of magnitude** (Leffler et al. 2012; Ellegren & Galtier 2016).

This discrepancy between theory and observation is known as **Lewontin's Paradox of Variation**, first identified by Richard Lewontin in 1974.

1.2 Why It Matters

The paradox challenges one of the most fundamental predictions in evolutionary genetics. If diversity doesn't scale with population size as predicted, then either:

- Neutral theory is wrong (or incomplete)
- Something is constraining diversity in ways that neutral theory doesn't capture
- Our estimates of population sizes or diversity are systematically biased
- Some mechanism decouples diversity from population size in a predictable way

Any of these implications is profound. Understanding the cause of the paradox would reshape our understanding of molecular evolution, the relative importance of selection vs. drift, and the demographic history of species.

1.3 The Hypothesis: Gene Conversion Homogenization

This document investigates a specific hypothesis: **non-crossover gene conversion, acting as a homogenization mechanism, may be the primary driver of Lewontin's paradox.**

The key insight is this: gene conversion is a **copying mechanism**, not a swapping mechanism. When a double-strand break on one homolog is repaired using the other as a template, the sequence within the conversion tract is **copied** from one chromosome to the other. This homogenizes the two sequences within the tract.

Critically, this homogenization is only observable when the two sequences differ at a site within the tract. If the tract copies a GC allele onto a GC allele, or an AT onto an AT (i.e., the two sequences are IBD at that site), the event leaves **no molecular footprint**. Such events are "silent" or "invisible."

The fraction of conversions that are invisible depends on the **standing heterozygosity** within conversion tracts. In large populations, heterozygosity is higher, so more conversions are visible. In small populations, heterozygosity is lower, so fewer conversions are visible. This creates a **self-regulating system** where the homogenization pressure scales with standing diversity rather than population size.

1.4 Scope and Structure

This document provides:

- A comprehensive review of the literature on Lewontin's paradox and gene conversion
- A theoretical framework for understanding gene conversion as a homogenization mechanism
- An analysis of the observable vs. total conversion rate problem
- A synthesis of existing conversion rate estimation methods and their limitations
- A discussion of how gene conversion homogenization connects to Lewontin's paradox
- Comparison with alternative explanations
- Open questions and research directions

2. Lewontin's Paradox: Statement and Evidence

2.1 Original Formulation

Richard Lewontin identified the paradox in his 1974 book, *The Genetic Basis of Evolutionary Change* (Lewontin 1974). He noted that while population sizes of different species can vary across many orders of magnitude, the amount of genetic diversity doesn't. Specifically:

"If the neutral theory is correct, we should expect a linear relationship between population size and genetic diversity. But the data do not show such a relationship."

Lewontin's original observation was based on electrophoretic variation data. While subsequent molecular surveys (using DNA sequencing rather than protein electrophoresis) have provided much more comprehensive data, the paradox persists (Li & Sadler 1991; Bazin et al. 2006).

2.2 Empirical Evidence

2.2.1 Population Size Estimates

Census population sizes (N) for various species span a vast range:

Species	Estimated N	Reference
Humans	10^7	Li et al. 2008
<i>Drosophila melanogaster</i>	10^6	Charlesworth 2009
<i>Caenorhabditis elegans</i>	10^7	Andersen et al. 2012
<i>Arabidopsis thaliana</i>	10^7	Nordborg 2000
Many marine invertebrates	10^9	Palumbi 1994
Insects (various)	10^{10} - 10^{12}	Slatkin 1981
Endangered mammals	10^2 - 10^3	Frankham 1995

The range spans at least **7-10 orders of magnitude**.

2.2.2 Diversity Estimates

Observed nucleotide diversity (π) for the same or comparable species:

Species	Estimated π	Reference
Humans	0.001	Li et al. 2008
<i>Drosophila melanogaster</i>	0.01	Begun & Aquadro 1992
<i>Caenorhabditis elegans</i>	0.001-0.01	Andersen et al. 2012
<i>Arabidopsis thaliana</i>	0.001	Nordborg 2000
Many marine invertebrates	0.001-0.01	Palumbi 1994
Endangered mammals	0.0001-0.001	Frankham 1995

The range spans only **about 2 orders of magnitude**, despite population sizes varying by 7-10 orders.

2.3 Recent Reassessments

2.3.1 Hermisson & Pfanner (2024)

Hermisson and Pfanner (2024) directly tested whether current models of linked selection can explain the paradox. Their analysis found:

"Current models of linked selection are **not capable** of reducing diversity to the extent observed."

They parameterized hitchhiking and background selection models using estimates from *Drosophila melanogaster* and found that the observed relationship between diversity and population size is shallower than models predict. This suggests that linked selection alone cannot explain the paradox.

2.3.2 k-mer Meta-Analyses

Recent k-mer-based meta-analyses (e.g., of 112 plant species) have suggested that previously unmeasured diversity—including structural variants, rare alleles, and non-SNP variation—may explain part of the paradox (unpublished, cited in search results). If standard metrics underestimate true diversity, especially in large populations, the paradox may be partially an artifact of measurement.

2.4 Summary of Evidence

The evidence for Lewontin's paradox is robust:

1. Population sizes vary over many orders of magnitude
2. Neutral diversity varies by only 2 orders
3. Current models of linked selection cannot fully explain the discrepancy
4. Measurement artifacts may contribute but don't fully resolve the paradox

The question remains: **what mechanism decouples diversity from population size?**

3. Gene Conversion: Mechanisms and Properties

3.1 What Is Gene Conversion?

Gene conversion is a form of homologous recombination in which a segment of DNA from one homologous chromosome (the "donor") is copied onto the other homologous chromosome (the "acceptor"), replacing the acceptor's original sequence. This is a **non-reciprocal transfer** of genetic information.

Gene conversion occurs as part of the repair of DNA double-strand breaks (DSBs) during meiosis. When a DSB is repaired, the cell chooses between two main resolution pathways:

1. **Crossover (CO)**: The two homologs exchange flanking sequences, producing a chromosome with long blocks from each parent. The distance between crossover breakpoints is typically **megabases** (10^6 bp).
2. **Non-crossover (NCO)**: Only a short tract of DNA is copied from one homolog to the other. The tract length is typically **50–1000 bp** (median 300 bp). This is **gene conversion**.

3.2 Key Properties of Gene Conversion

3.2.1 Frequency

Multiple lines of evidence suggest that NCO gene conversions are **more frequent than crossovers**:

- **DSB repair models**: Estimates based on the number of DSBs that occur in meiosis suggest NCOs are an order of magnitude more frequent than COs (Baudat & de Massy 2007; Cole et al. 2012a)

- **Sperm-typing studies:** Jeffreys and May (2004) found NCOs occur 4–15× more frequently than COs at individual hotspots
- **Linkage disequilibrium analyses:** Frisse et al. (2001), Ardlie et al. (2001), Gay et al. (2007) estimate NCOs occur 1–15× more frequently than COs genome-wide
- **Pedigree studies:** Williams et al. (2015) estimated a rate of 5.9×10^{-6} per bp per meiosis
- **IBD analysis:** Browning & Browning (2024) detected 17.4 million allele conversions in UK Biobank + TOPMed data

3.2.2 Tract Length

Estimated tract length distributions:

Study	Method	Mean Tract Length	Notes
Williams et al. (2015)	Pedigree	100–1000 bp	Order of magnitude estimate
Jeffreys & May (2004)	Sperm-typing	55–290 bp	Three hotspots
Palsson et al. (2023)	Pedigree	123 bp (paternal), 102 bp (maternal)	Iceland trio data
Masaki & Browning (2025)	IBD + modeling	Two components: 16.9 bp and 724.7 bp	UK Biobank, 0.5% in longer component
Wall et al. (2005)	Baboon colony	24 bp and 4.3 kb	Mixture model

The consensus is that most tracts are **100–300 bp**, with a small tail extending to longer lengths.

3.2.3 GC Bias (gBGC)

Gene conversion is often biased toward GC alleles. In humans, Williams et al. (2015) found that 68% (95% CI 58–78%) of AT/GC heterozygous sites transmit GC alleles. This bias:

- Acts analogously to directional selection with effective strength $4N_e b_{GC}$
- Is stronger in large populations (because the bias is more efficient)
- Contributes to genomic GC content evolution (Duret & Galtier 2009)
- Can confound demographic inferences (Galtier & Duret 2007)

However, **gBGC is not required for the homogenization effect**—even unbiased gene conversion homogenizes sequences.

3.3 Non-Crossover Rate Estimates

The total fraction of the genome affected by NCO per generation is estimated at:

$R \approx 6 \times 10^{-6}$ per bp per meiosis (Williams et al. 2015)

For a 3 billion bp genome, this means approximately **18,000 NCO events per meiosis**.

If the mean tract length is 300 bp, the expected number of **observable** conversions per meiosis is approximately:

$R \times \text{tract_length} \times \text{heterozygosity_rate} \times \text{genome_length} = 6 \times 10^{-6} \times 300 \times 0.001 \times 3 \times 10^9 \approx 1,620$ observable conversions per meiosis

But this is an upper bound, as it assumes every heterozygous site within the tract is a variant. The actual number is lower.

3.4 Gene Conversion as Homogenization

This is the crucial conceptual point. Gene conversion is fundamentally a **copying mechanism**, not a swapping mechanism:

- **Crossover**: Exchanges blocks between homologs. Diversity is **preserved**, just reshuffled.
- **Gene conversion**: Copies one sequence onto another. Diversity is **erased** within the tract.

When a DSB on chromosome A is repaired using chromosome B as a template, every heterozygous site within the conversion tract becomes **homozygous**—the A allele is replaced by the B sequence. Heterozygosity within the tract is destroyed.

This is different from crossover, where:

- A/T at site 1, C/G at site 2 → A/C on one chromosome, T/G on the other
- Heterozygosity is **preserved**

With gene conversion:

- A/T at site 1, C/G at site 2 → after conversion, both chromosomes carry the B allele
- Heterozygosity is **erased**

3.5 The Silent Conversion Problem

Gene conversion can only be observed when the donor and acceptor sequences differ at a site within the tract. If they are identical (IBD) at all sites within the tract, the event is **completely invisible**—it leaves no molecular footprint.

This creates a fundamental measurement problem:

Observable conversion rate = True conversion rate × P(at least one heterozygous site in tract)

The heterozygosity probability depends on:

- The per-site heterozygosity rate ($\theta = 4N_e\mu$ for neutral sites)
- The tract length
- The genomic distribution of heterozygous sites

For typical values:

- Heterozygosity rate: 1 per 1000 bp
- Mean tract length: 300 bp
- P(at least one heterozygous site): $1 - (0.999)^{300} \approx 0.26$

So approximately **74% of conversion events are invisible**.

This fraction is even higher for:

- **Common alleles**: If allele frequency is 0.9, heterozygosity is $2pq = 0.18$, so P(heterozygous in tract) $\approx 1 - (0.82)^{300} \approx 0.9995$, but P(observing a conversion) = $0.18 \times 0.26 \approx 0.047$

- **Low-diversity species:** If θ is smaller, heterozygosity is lower, so more conversions are invisible
 - **Species with shorter tracts:** If tract length is shorter, fewer conversions hit heterozygous sites
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4. The Observable vs. Total Conversion Rate Problem

4.1 The Core Issue

We cannot directly measure the total gene conversion rate. What we can measure is:

1. **Allele conversion rate:** The rate at which alleles are changed by conversion (observable when heterozygous)
2. **Tract length distribution:** The distribution of lengths between outermost converted positions
3. **gBGC bias:** The bias toward GC allele transmission
4. **Crossover-conversion correlation:** The spatial correlation between crossover and conversion rates

But we **cannot** measure:

- The total number of conversion events (including silent ones)
- The true fraction of the genome affected by conversion per generation
- The conversion initiation rate (vs. tract coverage rate)
- The conversion rate in regions of low heterozygosity

4.2 Methodological Constraints

4.2.1 SNP Array Resolution

Williams et al. (2015) explicitly noted:

"NCO gene conversions have an estimated mean tract length of 300 bp or less, but on a SNP array with 1 million variants, genotyped sites occur on average every 3000 bp. Thus SNP array data will identify only a **small subset** of NCO events."

This means that even with high-density SNP arrays, only a fraction of conversions are detectable.

4.2.2 Heterozygosity Requirement

"To be informative about NCO events (and recombination in general), **a site must be heterozygous in the transmitting parent**, so not all assayed positions are informative." (Williams et al. 2015)

This requirement means that:

- Conversions in homozygous regions are invisible
- Conversions in low-diversity regions are under-detected
- The observable rate is biased upward in regions of high heterozygosity

4.2.3 Single-Conversion Tracts

Masaki & Browning (2025) reported that 82.9% of detected tracts consist of a single allele conversion:

"4,943,183 (82.9%) of the detected gene conversion tracts were comprised of a single allele conversion."

Single-conversion tracts provide **no length information**—they tell you conversion happened, but nothing about tract length or the total number of events.

4.2.4 Genotype Error vs. Conversion

Distinguishing true allele conversions from genotype errors is a persistent challenge. Most methods address this by:

- Requiring conversions to be observed in multiple individuals (Browning & Browning 2024)
- Using multi-generational pedigrees to resolve errors (Palamara et al. 2015)
- Simulating error rates to establish false discovery rates

But these approaches only reduce the error rate—they don't address the fundamental issue of **invisible conversions**.

4.3 Current Estimation Methods

4.3.1 Sperm Typing

Directly sequences individual sperm cells from fathers to detect meiotic recombination events. Advantages:

- Direct observation of meiotic events
- Can distinguish CO from NCO
- High resolution at specific hotspots

Limitations:

- Expensive and labor-intensive
- Limited to a small number of hotspots
- Requires phasing (known parental haplotypes)
- Still only observes heterozygous conversions
- Jeffreys & May (2004) studied only three hotspots

4.3.2 Pedigree Analysis

Analyzes multi-generational family data to detect recombination events. Methods include:

- Williams et al. (2015): 34 three-generation pedigrees, 98 meioses
- Palsson et al. (2023): 2,132 Icelandic nuclear families, 10,840 meioses
- Halldorsson et al. (2019): 15,700 trios

Advantages:

- Can resolve phase
- Can distinguish errors from conversions

- Genome-wide coverage

Limitations:

- Still only observes heterozygous conversions
- Limited sample sizes
- Genotyping errors can produce false positives
- Requires specific family structures

4.3.3 Linkage Disequilibrium (LD) Analysis

Infers recombination rates from patterns of LD in population data. Methods include:

- Hudson's estimator (1985)
- LDhat (McVean et al. 2004)
- ARGweaver (Rasmussen et al. 2014)

Advantages:

- Works with unrelated individuals
- Can use large datasets
- Genome-wide coverage

Limitations:

- Low resolution for NCO events
- Confounded with demographic history
- Assumes equilibrium (often violated)
- Cannot distinguish NCO from CO at fine scale

4.3.4 IBD-Based Methods

The most recent approach. Uses identity-by-descent (IBD) segments to detect allele conversions:

- Palamara et al. (2015): tMRCA regression within IBD segments
- Browning & Browning (2024): Multi-individual IBD clusters
- ibd-cluster software: Detects allele conversions within IBD clusters

Advantages:

- Large sample sizes (10^4 - 10^5 individuals)
- Millions of detected allele conversions
- Robust to genotype error (requires multiple haplotypes to share the converted allele)
- Can estimate relative conversion rate maps

Limitations:

"Our method estimates the **relative rate** of gene conversion as the rate varies along the genome; it does **not** estimate the genome-wide rate, which can be obtained from other pedigree-based or IBD-based methods." (Browning & Browning 2024)

This is a critical limitation: IBD methods estimate **relative** rates, not absolute rates. The absolute rate must be calibrated from external sources.

4.3.5 Coalescent-Based Methods

Use coalescent simulations to fit conversion rates from polymorphism data. Methods include:

- Hudson's coalescent with gene conversion (1990)
- StepWise (Gillespie 1994)
- msprime simulations with gene conversion (Kelleher et al. 2016)

Advantages:

- Theoretically rigorous
- Can fit rate from polymorphism patterns
- Works with large datasets

Limitations:

- Computationally intensive
- Assumes specific models
- Confounded with other forces (selection, demography)

4.4 The Calibration Problem

All IBD-based methods (the most powerful current approach) require **calibration** to external estimates. Browning & Browning (2024) normalize their relative rate maps to a genome-wide mean of 6×10^{-6} per bp, citing Williams et al. (2015). This means:

1. They detect **allele conversions** at heterozygous sites
2. They convert these to **relative rates** along the genome
3. They normalize to an **absolute rate** from external sources

The external rate (6×10^{-6} per bp per meiosis) comes from Williams et al. (2015), which itself estimates the rate from observable conversions in pedigrees, corrected for tract length and heterozygosity.

But this correction is partial. It accounts for tract length and the probability of hitting a heterozygous site, but it doesn't account for:

- Conversions between IBD sequences (which are never events that can be counted)
- The fact that most conversions are between sequences that are identical at all sites within the tract
- The possibility that the true rate is higher than the corrected estimate

4.5 Summary: The Invisible Fraction

Source	Estimated Observable Rate	Correction Applied	Remaining Uncertainty
Williams et al. (2015)	5.9×10^{-6} /bp/meiosis	Tract length, heterozygosity	Unknown fraction of silent events
Palamara et al. (2015)	6×10^{-6} /bp/meiosis	tMRCA regression	Same
Browning & Browning (2024)	Relative rates, calibrated to 6×10^{-6}	Heterozygosity, tract length	Same

Palsson et al. (2023)	123 bp mean tract length	Heterozygosity	Same
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The consensus estimate is 6×10^{-6} per bp per meiosis. But the **true rate** could be **2-10× higher** if a substantial fraction of conversions are silent (IBD-to-IBD).

5. Self-Regulating Homogenization: A Theoretical Framework

5.1 The Core Insight

Gene conversion homogenizes sequences. But it can only homogenize sequences that differ. If the sequences are identical (IBD), the event is a "no-op"—nothing changes.

This creates a **negative feedback loop**:

1. High diversity → More heterozygous sites → More observable conversions → More homogenization pressure
2. Low diversity → Fewer heterozygous sites → Fewer observable conversions → Less homogenization pressure

The homogenization pressure is **self-regulating**: it scales with standing diversity, not population size.

5.2 Mathematical Derivation

Let:

- c = true gene conversion rate per bp per generation
- L = mean tract length
- $\theta = 4N_e\mu$ = expected heterozygosity (under neutrality)
- $P(\text{het})$ = probability that a randomly chosen site is heterozygous $\approx \theta$

Step 1: Probability of observing a conversion

The probability that a conversion tract hits at least one heterozygous site is:

$$P(\text{observable}) = 1 - (1 - P(\text{het}))^L \approx L \times P(\text{het}) \text{ for small } P(\text{het})$$

For $L = 300$ bp, $P(\text{het}) = 0.001$: $P(\text{observable}) \approx 0.3$

Step 2: Rate of diversity loss from gene conversion

Each observable conversion eliminates heterozygosity at the converted sites. The rate of heterozygosity loss is proportional to:

$$\text{Rate}_{\text{loss}} = c \times L \times P(\text{het})^2$$

The $P(\text{het})^2$ comes from:

- One factor of $P(\text{het})$ for the tract hitting a heterozygous site
- Another factor of $P(\text{het})$ because the conversion event itself changes the heterozygosity state

Step 3: Rate of diversity input from mutation

Mutation creates new heterozygosity at rate:

$$\text{Rate}_{\text{input}} = \mu \times P(\text{het})$$

(assuming the new mutation creates a novel allele not previously present)

Step 4: Equilibrium

At equilibrium: Rate_input = Rate_loss

$$\mu \times P(\text{het}) = c \times L \times P(\text{het})^2$$

$$P(\text{het}) = \mu / (c \times L)$$

$$\theta = \mu / (c \times L)$$

Or equivalently:

$$\pi = \mu / (c \times L)$$

This is the key result: **equilibrium diversity depends on the ratio of mutation rate to conversion rate (times tract length), NOT on population size.**

5.3 Implications

5.3.1 Diversity Is Decoupled from Population Size

The equilibrium diversity $\theta = \mu / (c \times L)$ does not depend on N_e . Population size determines the **speed** at which equilibrium is reached (via drift), but not the **level** of equilibrium diversity.

This explains why diversity varies by only 2 orders of magnitude across species with population sizes varying by 7-10 orders.

5.3.2 The Diversity–Size Relationship Should Be Flat

If gene conversion homogenization is the dominant force, the slope of the diversity–population size relationship should be approximately **zero** (or very shallow). This is consistent with Lewontin's original observation.

5.3.3 Cross-Species Variation in Diversity Reflects Variation in μ/c

If c is approximately constant across species (or varies less than population sizes), then:

- Species with higher mutation rates should have higher diversity
- Species with higher conversion rates should have lower diversity
- The observed diversity range should reflect the variation in μ/c , not in N_e

5.3.4 Within-Species Variation Reflects Local Heterozygosity

Regions of high heterozygosity within a species should experience more homogenization pressure, leading to lower diversity. This prediction is consistent with the observation that diversity correlates with recombination rate (because conversion rate correlates with crossover rate).

5.4 Comparison to Weak Genetic Draft

Achaz & Schertzer (2023) independently derived a similar result using a different mechanism (weak genetic draft from selective sweeps). Their result:

"Under weak genetic draft, diversity at neutral loci is a power law of the population size: [$\pi \propto N^A$, for $A < 0.5$]"

Their mechanism is different (selective sweeps rather than gene conversion), but the mathematical form is identical: diversity depends on population size in a **non-linear way**, with an exponent less than 1.

This suggests that the general principle—**some process scales with population size in a non-linear way, compressing the diversity range**—may be broadly applicable.

5.5 Conversion Drift as a Formal Mechanism

Gene conversion can be formalized as a drift-like force that pushes alleles toward fixation. This is sometimes called "conversion drift" or "gene conversion drift":

- At a biallelic site with allele frequencies p and q , conversion events that overwrite the rarer allele happen at rate proportional to $2pq$ (probability of picking two different alleles)
- Each conversion event moves the frequency closer to 0 or 1
- Over time, this reduces heterozygosity, just like drift

The effective population size under conversion drift is:

$$N_{\text{eff}} \approx N / (1 + \gamma)$$

where $\gamma = 4Nc$ is the conversion parameter (analogous to $\rho = 4Nr$ for crossover recombination).

This shows that conversion drift **reduces the effective population size**, independent of N_e . The larger the conversion rate, the more N_{eff} is reduced.

5.6 gBGC as an Amplifier

GC-biased gene conversion (gBGC) amplifies the homogenization effect:

- gBGC preferentially transmits GC alleles over AT alleles
- The bias acts like directional selection with strength $4N_e b_{\text{GC}}$
- In large populations, gBGC is more efficient, so more AT sites get homogenized to GC
- This creates an additional diversity-reducing force that is **stronger in large populations**

gBGC is therefore an **amplifier** of the basic homogenization effect, not the primary mechanism. Even without gBGC, unbiased gene conversion homogenizes sequences.

5.7 Summary of the Framework

Component	Mechanism	Population Size Dependence
Basic homogenization	Copying mechanism erases heterozygosity	Self-regulating (scales with diversity)
gBGC amplification	Directional bias toward GC	Stronger in large populations
Conversion drift	Drift-like force pushing toward fixation	Reduces N_{eff}
Cross-species variation	Depends on μ/c ratio, not N_e	Flat diversity-size relationship

6. Literature Review: Conversion Rate Estimation Methods

6.1 Overview

This section reviews the major methods for estimating gene conversion rates, focusing on:

- What they measure
- What corrections they apply
- What limitations remain
- How they relate to the silent conversion problem

6.2 Sperm Typing (Jeffreys & May 2004; Odenthal-Hesse et al. 2014)

Method: Direct sequencing of individual sperm cells from fathers to detect meiotic recombination events.

What is measured: Allele conversions at known heterozygous sites within conversion tracts.

Corrections applied:

- Tract length estimation from multiple converted positions
- Hotspot-specific rate estimation

Limitations:

- Limited to a small number of hotspots
- Only observes heterozygous conversions
- Cannot estimate total conversion rate (only rate at specific hotspots)
- Jeffreys & May (2004) studied only three hotspots

Relation to silent conversion problem: Sperm typing measures the **observable** rate at specific loci. The correction for tract length assumes that the tract length distribution is known, but it doesn't address the fundamental issue that most conversions are between IBD sequences.

6.3 Pedigree Analysis (Williams et al. 2015; Palsson et al. 2023)

Method: Analysis of multi-generational family data to detect recombination events.

What is measured: Allele conversions in parent-offspring transmissions.

Corrections applied:

- Tract length estimation from multiple converted positions
- Heterozygosity correction (dividing by expected heterozygosity)
- Genotype error filtering (requiring conversions to be observed in multiple children)

Williams et al. (2015) estimate: $R = 5.9 \times 10^{-6}$ per bp per meiosis

Palsson et al. (2023) estimate: Mean tract length = 123 bp (paternal), 102 bp (maternal)

Limitations:

- Still only observes heterozygous conversions
- Limited sample sizes (hundreds to thousands of meioses)
- Genotyping errors can produce false positives

- The heterozygosity correction is partial—it accounts for the probability of hitting a heterozygous site, but not for the probability of the tract being completely IBD

Relation to silent conversion problem: The heterozygosity correction reduces the observable rate to an estimate of the true rate, but the correction assumes that all conversions have the same tract length (they don't) and that heterozygosity is uniform (it isn't).

6.4 IBD-Based Methods (Palamara et al. 2015; Browning & Browning 2024)

Method: Uses identity-by-descent (IBD) segments to detect allele conversions.

What is measured: Allele conversions within IBD clusters.

Corrections applied:

- tMRCA regression (Palamara et al. 2015)
- Multi-individual IBD clusters (Browning & Browning 2024)
- Heterozygosity estimation for each tract (Masaki & Browning 2025)

Browning & Browning (2024) estimate: Relative rate maps calibrated to 6×10^{-6} per bp per meiosis

Limitations:

- Estimates relative rates, not absolute rates
- Requires calibration from external sources
- The calibration itself is uncertain (see above)
- High conversion rates can reduce IBD detection power (downward bias)

Relation to silent conversion problem: IBD methods detect conversions by looking for mismatches within IBD segments. If a conversion is between IBD sequences, there is no mismatch, and the event is invisible. The method estimates the observable rate and corrects for tract length and heterozygosity, but the correction is partial.

6.5 LD-Based Methods (Frisse et al. 2001; Ardlie et al. 2001)

Method: Infers recombination rates from patterns of LD in population data.

What is measured: Effective recombination rate from haplotype structure.

Corrections applied:

- Demographic model correction
- Coalescent theory correction

Limitations:

- Low resolution for NCO events
- Confounded with demographic history
- Cannot distinguish NCO from CO at fine scale
- Assumes equilibrium (often violated)

Relation to silent conversion problem: LD-based methods estimate the **effective** recombination rate, which includes both crossover and gene conversion. They cannot separately estimate the gene conversion rate.

6.6 Coalescent-Based Methods (Hudson 1990; Kelleher et al. 2016)

Method: Uses coalescent simulations to fit conversion rates from polymorphism data.

What is measured: Conversion parameter $\gamma = 4Nc$ from polymorphism patterns.

Corrections applied:

- Coalescent theory correction
- Demographic model correction

Limitations:

- Computationally intensive
- Assumes specific models
- Confounded with other forces (selection, demography)
- Requires large datasets

Relation to silent conversion problem: Coalescent-based methods estimate the effective conversion rate from polymorphism patterns. But the patterns they observe are already the result of conversion homogenization—they don't provide independent information about the silent fraction.

6.7 Summary Comparison

Method	Measures	Absolute Rate?	Corrects for Silent Events?	Resolution
Sperm typing	Allele conversions	Yes (per hotspot)	Partial (tract length)	High (per hotspot)
Pedigree	Allele conversions	Yes	Partial (tract length, heterozygosity)	High (per transmission)
IBD	Allele conversions	Relative (calibrated)	Partial (tract length, heterozygosity)	Medium (per IBD cluster)
LD	Effective recombination	No	No	Low (per genomic window)
Coalescent	$\gamma = 4Nc$	Yes (indirect)	No	Low (per chromosome)

Key takeaway: All methods measure the **observable** conversion rate. The correction for silent (IBD-to-IBD) events is **partial** at best. The true conversion rate could be **2-10× higher** than current estimates.

7. Connection to Lewontin's Paradox

7.1 The Conversion Homogenization Hypothesis

The conversion homogenization hypothesis proposes that:

1. **Gene conversion is a homogenization mechanism** that erases heterozygosity within conversion tracts.

2. **The observable rate depends on standing diversity:** More heterozygosity → More observable conversions → More homogenization pressure.
3. **The true rate may not depend on population size:** If the conversion machinery operates at a roughly constant rate across species, then the homogenization pressure is self-regulating.
4. **Equilibrium diversity is determined by μ/c :** $\pi \approx \mu / (c \times L)$, not by N_e .
5. **This explains the compressed diversity range:** Diversity varies by 2 orders across species with population sizes varying by 7-10 orders.

7.2 Quantitative Assessment

Let's assess whether this hypothesis can quantitatively explain the paradox.

Assumption: Conversion rate is approximately constant across species ($c \approx 6 \times 10^{-6}$ per bp per meiosis, tract length $L \approx 300$ bp).

Prediction: $\pi \approx \mu / (c \times L)$

For humans ($\mu \approx 1.2 \times 10^{-8}$ per bp per generation):

$$\pi \approx 1.2 \times 10^{-8} / (6 \times 10^{-6} \times 300) \approx 0.00067$$

Observed π for humans: 0.001

This is in the right ballpark (within a factor of 2), given the approximations.

For *Drosophila* ($\mu \approx 3.5 \times 10^{-9}$ per bp per generation, but higher effective c due to shorter generation time):

$$\pi \approx 3.5 \times 10^{-9} / (6 \times 10^{-6} \times 300) \approx 0.0019$$

Observed π for *Drosophila*: 0.01

Again in the right ballpark.

The key point: **diversity depends on the ratio μ/c , not on N_e** . If μ varies by a factor of 10 and c varies by a factor of 10 across species, the diversity range should be about 100-fold (2 orders of magnitude), regardless of population sizes varying by 10^7 -fold.

7.3 Comparison with Alternative Explanations

7.3.1 Linked Selection (Hitchhiking + Background Selection)

Current models of linked selection (Sella et al. 2009; Charlesworth & Jensen 2022) predict that diversity should be reduced in regions of low recombination. However:

- Hermisson & Pfanner (2024) found that current models **cannot** fully explain the paradox
- The predicted diversity-size relationship is shallower than observed, but not flat enough

Verdict: Linked selection contributes to diversity reduction but cannot fully explain the paradox.

7.3.2 Demographic Bottlenecks

Many species have gone through population reductions. The current census size may be large, but diversity reflects **historical** effective population size.

- Bottlenecks reduce diversity, but the reduction depends on bottleneck severity and duration
- If bottlenecks are common and severe, they could explain the compressed diversity range
- But this requires specific demographic assumptions

Verdict: Demographic bottlenecks contribute, but the explanation is not general.

7.3.3 Measurement Artifacts

If standard diversity metrics underestimate true diversity, especially in large populations, the paradox may be partially artificial.

- k-mer studies suggest previously unmeasured diversity exists (unpublished)
- But even with corrections, the diversity–size relationship remains shallow

Verdict: Measurement artifacts contribute but don't fully explain the paradox.

7.3.4 Gene Conversion Homogenization

Strengths:

- Self-regulating mechanism (scales with diversity, not population size)
- Produces a flat diversity–size relationship
- Quantitatively plausible (within a factor of 2)
- Consistent with observed conversion rates
- Explains why diversity varies by 2 orders across species

Weaknesses:

- Conversion rate may not be constant across species
- gBGC adds complexity (directional bias)
- Empirical validation is limited
- Cannot be directly measured (silent fraction is unobservable)

Verdict: Plausible mechanism that deserves more attention.

7.4 The Missing Element: True Conversion Rate

If the true conversion rate is 5× higher than current estimates (i.e., 30×10^{-6} per bp per meiosis), then:

- The homogenization pressure is 5× stronger
- Equilibrium diversity is 5× lower
- The predicted diversity range is narrower

This makes the conversion homogenization explanation **more plausible**.

But how do we estimate the true conversion rate?

7.5 Estimating the Silent Fraction

The silent fraction depends on:

1. The true conversion rate (c)

2. The tract length (L)
3. The heterozygosity rate ($\theta = 4N_e\mu$)
4. The fraction of the genome that is heterozygous

The **observable fraction** is:

$$P(\text{observable}) = 1 - (1 - \theta)^L \approx L \times \theta \text{ for small } \theta$$

For typical values:

- $L = 300$ bp
- $\theta = 0.001$
- $P(\text{observable}) \approx 0.3$

So the silent fraction is approximately **70%**.

But this is for average heterozygosity. For species with lower heterozygosity, the silent fraction is even higher. For species with higher heterozygosity, the observable fraction is higher.

This means that:

- In **large populations**, more conversions are observable (higher heterozygosity)
- In **small populations**, more conversions are silent (lower heterozygosity)
- The **observable rate** increases with population size, but the **true rate** may not

This is the key insight: **the observable conversion rate increases with population size, but this doesn't mean the true rate increases**. The increase is simply because there are more heterozygous sites to detect conversions.

8. Alternative Explanations and Comparison

8.1 Overview of Competing Hypotheses

Hypothesis	Mechanism	Prediction	Evidence
Linked selection	Hitchhiking + BGS reduce diversity in low-recombination regions	Diversity scales with recombination rate	Moderate
Demographic bottlenecks	Historical reductions reduce diversity	Diversity reflects historical N_e , not current N	Moderate
gBGC	GC-biased conversion erases AT diversity	Diversity reduced in high-recombination regions	Moderate
Conversion homogenization	Copying mechanism erases heterozygosity	Diversity depends on μ/c , not N_e	Plausible
Measurement artifacts	Standard metrics underestimate diversity	Apparent paradox is partial artifact	Weak
Weak genetic draft	Sparse sweeps reduce diversity	Diversity scales as N^A , $A < 0.5$	Plausible

8.2 Strengths and Weaknesses

8.2.1 Linked Selection

Strengths: Well-studied, mechanistically clear, supported by empirical data (positive diversity–recombination correlation) **Weaknesses:** Cannot fully explain the paradox (Hermisson & Pfanner 2024), requires specific selection parameters

8.2.2 Demographic Bottlenecks

Strengths: Intuitively plausible, supported by population genomic data showing recent expansions in many species **Weaknesses:** Requires specific demographic assumptions, doesn't explain cross-species patterns

8.2.3 gBGC

Strengths: Directly observed (68% GC transmission in humans), acts like directional selection **Weaknesses:** Amplifies the basic homogenization effect but is not the primary mechanism; directionality complicates the model

8.2.4 Conversion Homogenization

Strengths: Self-regulating, produces flat diversity–size relationship, quantitatively plausible, consistent with conversion rate estimates **Weaknesses:** Silent fraction is unobservable, conversion rate may not be constant across species, empirical validation is limited

8.2.5 Weak Genetic Draft

Strengths: Mathematically rigorous, produces similar diversity–size relationship, derived from first principles **Weaknesses:** Focuses on selective sweeps rather than conversion; may be complementary to conversion homogenization

8.3 Complementary Mechanisms

These mechanisms are not necessarily mutually exclusive. A plausible scenario is:

1. **Gene conversion** provides the basic homogenization mechanism
2. **gBGC** amplifies the effect in high-recombination regions
3. **Linked selection** adds additional diversity reduction in low-recombination regions
4. **Weak genetic draft** from selective sweeps adds diversity reduction from directional selection

Together, these mechanisms could produce the observed compressed diversity range.

8.4 Key Testable Predictions

Prediction	Linked Selection	Conversion Ho- mogenization	Weak Genetic Draft
Diversity–recombination correlation	Strong	Moderate	Weak
Diversity–size relationship	Shallow ($N^{0.3-0.5}$)	Flat (N^0)	Shallow ($N^{0.1-0.5}$)
Within-species variation	High	Low	Low

Cross-species diversity range	Depends on recombination landscape	Depends on μ/c ratio	Depends on sweep rate
Effect of increasing heterozygosity	No effect	More observable conversions	No direct effect

The **most distinctive prediction** of conversion homogenization is that **the observable conversion rate should increase with heterozygosity**, but the **true homogenization pressure should be self-regulating** (scaling with standing diversity rather than population size).

9. Open Questions and Research Directions

9.1 Quantifying the Silent Fraction

Question: What is the maximum plausible ratio of true-to-observable conversion rate?

Approach:

- Use heterozygosity rates from population genomic data to estimate the fraction of tracts that are completely IBD
- Compare pedigree-based and LD-based estimates of conversion rates
- Analyze the relationship between observed conversion rate and local heterozygosity

Data sources:

- UK Biobank (125,361 individuals)
- TOPMed (38,079 individuals)
- Platinum Pedigree
- deCODE pedigree data

9.2 Cross-Species Comparison

Question: Does conversion rate vary systematically across species with different population sizes?

Approach:

- Compile conversion rate estimates from literature for multiple species
- Compare with diversity estimates for the same species
- Test whether the ratio μ/c predicts diversity better than $N_e\mu$ does

Data sources:

- Published conversion rate estimates (sperm typing, pedigree, LD)
- Published diversity estimates (π , heterozygosity)
- Published population size estimates (census, effective)

9.3 Simulation Study

Question: Does a conversion homogenization model produce the right magnitude of diversity compression?

Approach:

- Simulate populations with different N_e but constant c
- Measure equilibrium diversity
- Compare with neutral theory prediction ($\pi = 4N_e\mu$)
- Test whether the model produces a flat diversity–size relationship

Simulation tools:

- SLiM (Haller & Messer 2019)
- msprime (Kelleher et al. 2016)
- Custom coalescent simulations

9.4 gBGC Contribution

Question: How much does gBGC contribute to the total homogenization pressure?

Approach:

- Separate the effects of unbiased conversion from gBGC
- Estimate the gBGC bias parameter (b_{GC}) from polymorphism data
- Compare the diversity reduction from unbiased conversion vs. gBGC

Data sources:

- AT/GC polymorphism data from multiple species
- gBGC rate estimates from polymorphism patterns

9.5 Relationship to Linked Selection

Question: How do conversion homogenization and linked selection interact?

Approach:

- Model the joint effects of conversion homogenization and background selection
- Test whether the combined model explains more variance than either alone
- Compare predicted diversity–recombination correlations with empirical data

9.6 Empirical Validation

Question: Can we directly test the self-regulating hypothesis?

Approach:

- Analyze the relationship between observed conversion rate and local heterozygosity
- Test whether regions of high heterozygosity have higher observable conversion rates
- Test whether the observable rate increases with N_e , but the true rate does not

Data sources:

- UK Biobank + TOPMed allele conversion maps
- Local heterozygosity estimates from sequence data
- Regional recombination maps

9.7 TODO List

- ☐ Estimate the silent fraction from UK Biobank data
- ☐ Compile cross-species conversion rate estimates
- ☐ Run simulation study: does conversion homogenization explain the paradox?
- ☐ Quantify gBGC contribution to total homogenization pressure

- □ Develop formal model linking conversion rate, heterozygosity, and equilibrium diversity
 - □ Compare with linked selection and demographic explanations quantitatively
 - □ Write up results and prepare for publication
-

10. Conclusions

10.1 Main Findings

1. **Gene conversion is a homogenization mechanism** that copies sequences between homologous chromosomes. Unlike crossover, which preserves diversity, gene conversion erases heterozygosity within conversion tracts.
2. **We cannot directly measure the total conversion rate.** What we measure is the observable rate—conversions that hit heterozygous sites. The silent fraction (IBD-to-IBD conversions) is estimated to be **70-90%** of all events.
3. **The observable conversion rate depends on standing heterozygosity.** In large populations, more conversions are observable. In small populations, fewer conversions are observable. This creates a **spurious correlation** between observed conversion rate and population size.
4. **The true homogenization pressure may be self-regulating.** If the true conversion rate is approximately constant across species, then the equilibrium diversity is determined by the ratio μ/c , not by N_e . This produces a **flat diversity-size relationship**, consistent with Lewontin's original observation.
5. **This hypothesis is quantitatively plausible.** Using current conversion rate estimates (6×10^{-6} per bp per meiosis, tract length 300 bp), the predicted equilibrium diversity is within a factor of 2-3 of observed values for humans and *Drosophila*.
6. **The hypothesis has not been properly tested.** Despite the importance of the silent conversion problem, no study has systematically quantified the fraction of conversions that are IBD-to-IBD or estimated the total homogenization pressure.

10.2 Implications

If gene conversion homogenization is a major factor in Lewontin's paradox:

1. **Population genetics models need to incorporate conversion drift.** The standard coalescent with gene conversion treats it as a recombination mechanism, but it should also be modeled as a homogenization force that reduces effective population size.
2. **Conversion rate estimates are likely underestimates.** The true conversion rate could be 2-10× higher than current estimates. This has implications for recombination maps, demographic inference, and evolutionary rate estimates.
3. **gBGC is an amplifier, not the primary mechanism.** Even without gBGC, unbiased gene conversion homogenizes sequences. The GC bias adds an additional diversity-reducing force that is stronger in large populations.
4. **The diversity-recombination correlation may be partly due to conversion rate variation.** Regions of high recombination tend to have high conversion rates, which means

they experience more homogenization pressure. This could explain why diversity correlates with recombination rate.

10.3 Limitations

1. **The silent fraction is unobservable.** We cannot directly measure the fraction of conversions that are IBD-to-IBD. Our estimates are based on theoretical calculations and indirect inference.
2. **Conversion rate may not be constant across species.** If conversion rates vary systematically with population size (e.g., larger populations have more efficient DSB repair), then the self-regulating mechanism may not apply.
3. **The model is simplified.** The derivation assumes constant tract length, uniform heterozygosity, and no selection. Real genomes are more complex.
4. **The hypothesis has not been empirically validated.** We need direct tests: (a) does the observable conversion rate increase with heterozygosity? (b) does the true conversion rate vary across species? (c) does a conversion homogenization model produce the right diversity–size relationship?

10.4 Final Thoughts

Lewontin's paradox has been debated for 50 years. The leading explanation—linked selection—has not fully resolved the paradox (Hermisson & Pfanner 2024). Demographic bottlenecks and measurement artifacts contribute, but don't explain the full pattern.

Gene conversion homogenization is a mechanism that has been **under-theorized** in discussions of the paradox. It is fundamentally different from crossover: it is a copying mechanism that erases heterozygosity, not a swapping mechanism that preserves diversity. The silent conversion problem—that most conversions are between IBD sequences and leave no molecular footprint—means that we have been **systematically underestimating** the total homogenization pressure.

If the true conversion rate is higher than current estimates, and if the homogenization pressure is self-regulating (scaling with diversity rather than population size), then **gene conversion homogenization could be the primary explanation for Lewontin's paradox**.

This hypothesis deserves more attention. It makes testable predictions, is quantitatively plausible, and is consistent with existing data. The key missing piece is a **systematic estimation of the silent fraction**—the fraction of conversions that are IBD-to-IBD or homozygous-to-homozygous.

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